rt here ✕ marix operation.c ✕

```c
#include <stdio.h>
void readMatrix(int a[10][10], int r, int c) {
    int i, j;
    for(i = 0; i < r; i++) {
        for(j = 0; j < c; j++) {
            scanf("%d", &a[i][j]);
        }
    }
}

void displayMatrix(int a[10][10], int r, int c) {
    int i, j;
    for(i = 0; i < r; i++) {
        for(j = 0; j < c; j++) {
            printf("%d\t", a[i][j]);
        }
        printf("\n");
    }
}

void addMatrix(int a[10][10], int b[10][10], int r, int c) {
    int sum[10][10], i, j;

    for(i = 0; i < r; i++) {
```

```c
        for(j = 0; j < c; j++) {
            sum[i][j] = a[i][j] + b[i][j];
        }
    }

    printf("\nMatrix Addition:\n");
    displayMatrix(sum, r, c);
}

void multiplyMatrix(int a[10][10], int b[10][10], int r1, int c1, int c2) {
    int mul[10][10], i, j, k;

    for(i = 0; i < r1; i++) {
        for(j = 0; j < c2; j++) {
            mul[i][j] = 0;
            for(k = 0; k < c1; k++) {
                mul[i][j] += a[i][k] * b[k][j];
            }
        }
    }

    printf("\nMatrix Multiplication:\n");
    displayMatrix(mul, r1, c2);
}
```

rt here matrix operation.c

```c

void transposeMatrix(int a[10][10], int r, int c) {
    int trans[10][10], i, j;

    for(i = 0; i < r; i++) {
        for(j = 0; j < c; j++) {
            trans[j][i] = a[i][j];
        }
    }

    printf("\nTranspose of Matrix:\n");
    displayMatrix(trans, c, r);
}

int main() {
    int a[10][10], b[10][10];
    int r, c;

    printf("Enter rows and columns of matrices: ");
    scanf("%d%d", &r, &c);

    printf("Enter elements of Matrix A:\n");
    readMatrix(a, r, c);

```

rt here ✕ marix operation.c ✕

```c
    printf("\nTranspose of Matrix:\n");
    displayMatrix(trans, c, r);
}

int main() {
    int a[10][10], b[10][10];
    int r, c;

    printf("Enter rows and columns of matrices: ");
    scanf("%d%d", &r, &c);

    printf("Enter elements of Matrix A:\n");
    readMatrix(a, r, c);

    printf("Enter elements of Matrix B:\n");
    readMatrix(b, r, c);

    addMatrix(a, b, r, c);
    multiplyMatrix(a, b, r, c, c);
    transposeMatrix(a, r, c);

    return 0;
}
```

```
Enter rows and columns of matrices: 2 2
Enter elements of Matrix A:
1 2
3 4
Enter elements of Matrix B:
5 6
7 8

Matrix Addition:
6       8
10      12

Matrix Multiplication:
19      22
43      50

Transpose of Matrix:
1       3
2       4

Process returned 0 (0x0)   execution time : 41.919 s
Press any key to continue.
```